

Yusuf Erdem

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PROFILE

Computer Engineering graduate (AYBU) and M.Sc. AI Engineering student at the University of Passau. Hands-on experience across SAP logistics IT, applied ML, and multi-agent AI systems.

EDUCATION

M.Sc. Artificial Intelligence Engineering 10/2026 - Present

University of Passau — Passau, Germany

B.Sc. Computer Engineering Graduated 2026

Ankara Yıldırım Beyazıt University (AYBU) — Ankara, Turkey

GPA: 3.13 / 4.0 (2.3 German scale)

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks, NLP, Probability & Statistics, Neural Networks

EXPERIENCE

AI Department Intern, Martur Fompak International

08/2026 - 09/2026

Bursa, Turkey

- Built a multi-agent AI system on Google's Agent Development Kit (ADK): a Gemini-based orchestrator coordinating knowledge-base search, parsing, and automated editing tools, with document edits delegated to Groq-hosted LLMs
- Connected an independently-built remote agent via the Agent-to-Agent (A2A) protocol, achieving full bidirectional cross-machine agent communication
- Independently designed a self-correcting Orchestrator/Editor/Reviewer review loop using ADK's LoopAgent, applying a reflection-style QA pattern (see Agent Council project)

IT Intern, SAP Logistics, Martur Fompak International

07/2026 - 08/2026

Bursa, Turkey

- Worked within a SAP-based Logistics IT team resolving factory workflow issues, shadowing ABAP screen development
- Contributed to a Retrieval-Augmented Generation (RAG) chatbot for the Sales module, converting internal documentation into a structured knowledge base

Data Science Intern, Acun Medya Akademi

01/2025 - 03/2025

Remote

- Analyzed large-scale datasets using Python, SQL, and data visualization libraries (Pandas, Matplotlib, Seaborn)
- Applied data preprocessing, feature engineering, and statistical analysis methods to support business decisions

Software Developer, Belsis (Post-Internship)

09/2023 - 04/2024

Ankara, Turkey

- Engineered custom Odoo ERP modules for inventory, sales, and BI reporting in Python, deployed to production supporting 50+ concurrent users
- Managed PostgreSQL databases with pgAdmin/DBBeaver, and used Git, GitLab, and Jira for team collaboration

Software Developer Intern, Belsis

06/2023 - 09/2023

Ankara, Turkey

- Designed a parcel-selection GIS application with OpenLayers, adopted by multiple Turkish municipalities
- Built the backend with C#, ASP.NET Core, and PostgreSQL+PostGIS, and the frontend with Node.js, JavaScript, HTML, and CSS

PROJECTS

AI-Based Health Risk Scoring & Resource Allocation

2025 – 2026

(Capstone) Graduation Project

- Co-developed a 3-person AI-based early warning and hospital-routing system for diabetes, hypertension, and stroke using exclusively non-laboratory, self-reportable patient data
- Owned the hypertension risk model end-to-end: removed data-leakage features, selected Logistic Regression (~82%) over an overfitting-prone XGBoost baseline (~98%), prioritizing generalizability and clinical interpretability
- Designed the three-tier (Green/Yellow/Red) risk stratification system and the overall pipeline integrating the disease models, hospital-routing optimizer, and web interface
- Awarded a TÜBİTAK 2209-A national undergraduate research grant; presented at MUIISC 2026 (Maltepe University)

Tech: Python, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Agent Council: Multi-Agent Vault Manager

2026

AI Engineering Project

- Built a multi-agent system on Google's Agent Development Kit that autonomously maintains a markdown knowledge base: an Orchestrator (Gemini) plans edits, an Editor (Groq) executes rewrites, and a Reviewer validates the result
- Implemented a self-correcting review loop using ADK's LoopAgent, with each agent given a distinct role so the system revises its own output until an independent reviewer approves it

Tech: Python, Google ADK, Gemini, Groq API

End-to-End Crypto Forecaster (Live in Production)

2025 – Present

Personal Project

- Zero-infrastructure MLOps pipeline: GitHub Actions cron jobs, file-based data store, idempotent ETL, and automated weekly model retraining
- Compared SARIMAX (1.8% MAPE) vs Bi-LSTM (3.2% MAPE), proving statistical models outperform deep learning on high-noise cryptocurrency data
- Deployed a live Streamlit dashboard for real-time BTC price forecasting, accessible at btcforecaster.streamlit.app

Tech: Python, TensorFlow, scikit-learn, GitHub Actions, Streamlit

SKILLS

Machine Learning: XGBoost, TensorFlow, scikit-learn, PyTorch, Feature Engineering

Agentic AI: Google ADK, Multi-Agent Systems, RAG, MCP, A2A Protocol, LangGraph

Data Engineering: PostgreSQL, Docker, ETL Pipelines, GitHub Actions CI/CD

Backend: FastAPI, Flask, ASP.NET Core, RESTful APIs

NLP: FastText, Sentiment Analysis, Text Classification

CERTIFICATIONS & AWARDS

- IELTS Academic — Band 8 (C1 Proficient Level)
- Complete A.I. & Machine Learning, Data Science Bootcamp — Udemy (2025)
- “Most Ecological Project” Award — 2024 UDHAM National Universities Idea Competition | 50,000 TL grant recipient

LANGUAGES

- Turkish: Native
- English: Full Professional Proficiency (IELTS Band 8, C1)
- German: A2 (currently completing B1 coursework)